# Expected Payoff KEY

Name _____

Is a lottery ticket worth buying? Is a protection plan worth the money? The expected value of the **payoff** tells you what happens on average.

| |
|---|
| **Net gain** = what you win − what you paid |
| **Expected payoff** = $\Sigma$ (net gain $\cdot$ probability) |

**Example 1: a lottery ticket.** A ticket costs \$2. 1 in 1,000 tickets wins \$500. 1 in 50 tickets wins \$10. The rest win nothing.

| Outcome | Net gain $x$ | $P(x)$ | $x \cdot P(x)$ |
|---|---|---|---|
| Win \$500 | $500 - 2 = 498$ | 0.001 | 0.498 |
| Win \$10 | $10 - 2 = 8$ | 0.02 | 0.16 |
| Win nothing | $-2$ | 0.979 | $-1.958$ |
| | $E(X) = \Sigma\, x \cdot P(x)$ | | $-1.30$ |

On average, a buyer **loses \$1.30** per ticket. The lottery keeps that money.

**Example 2: a protection plan.** A phone costs \$300 to replace. The chance it breaks this year is 0.10. A protection plan costs \$40 for the year.

Expected cost with no plan: $0.10 \cdot 300 = \$30$
Cost of the plan: \$40

On average, the plan costs \$10 more than paying for a broken phone yourself. That is how the company makes money.

**But:** the plan protects you from a sudden \$300 bill. Expected value is one part of a decision. How much risk you can afford is another.

Work with a partner. Compare answers when you finish each problem.

1. A school sells 500 raffle tickets for \$5 each. One ticket wins \$1,000.

   (a) Fill in the table.

| Outcome | Net gain $x$ | $P(x)$ | $x \cdot P(x)$ |
|---|---|---|---|
| Win | 995 | $\frac{1}{500}$ | 1.99 |
| Lose | $-5$ | $\frac{499}{500}$ | $-4.99$ |

   (b) Find the expected payoff. What does it mean? $1.99 - 4.99 = -\$3$. On average, a buyer loses \$3 per ticket.

2. A game costs \$1 to play. You spin a spinner: half the time you win nothing, a quarter of the time you win \$1, and a quarter of the time you win \$3.

   (a) Find the expected payoff.
   Net gains: $-1$, 0, 2. $(-1)(0.5) + 0(0.25) + 2(0.25) = -0.5 + 0 + 0.5 = 0$

   (b) Is the game fair? Yes. The expected payoff is 0.

3. At a carnival, you pay \$2 and draw one card from a deck. An ace wins \$10. A face card (J, Q, K) wins \$5. Anything else wins nothing.

   (a) Find the probability of each outcome. Ace $\frac{4}{52}$, face card $\frac{12}{52}$, other $\frac{36}{52}$

   (b) Find the expected payoff. Should you expect to win money?
   $8 \cdot \frac{4}{52} + 3 \cdot \frac{12}{52} + (-2) \cdot \frac{36}{52} = \frac{32 + 36 - 72}{52} = -\frac{4}{52} \approx -\$0.08$
   No. You lose about 8 cents a game on average.

4. A laptop repair costs \$400. The chance a laptop needs repair in its first year is 0.15. A one-year warranty costs \$75.

   (a) Find the expected repair cost without the warranty. $0.15 \cdot 400 = \$60$

   (b) Which is cheaper on average: buying the warranty or not? Not buying it (\$60 is less than \$75).

   (c) Give a reason someone might still buy the warranty.
   They could not afford a sudden \$400 bill. The warranty removes that risk.

5. The store sells 1,000 of these warranties. About how much money does the store expect to make? Explain.

   It collects $1{,}000 \cdot 75 = \$75{,}000$. About 150 laptops need repair: $150 \cdot 400 = \$60{,}000$. It expects to make about \$15,000.